Q1. Data Quality Assessment and Cleaning

Identification of Missing Values:

The data set was checked to identify the missing values. The basic information variables, such as Player, Country, Team, Age, Captaincy Experience, and Playing Role, did not have missing values, which means no processing was needed for these variables.

Most of the missing values were identified in the batting statistics, bowling statistics, and T20 performance statistics variables. These variables, such as Inns, Runs, BF, Average, SR, B Inns, B Wickets, etc., had missing values.

A detailed analysis of the data revealed that the missing values in the data set were not due to any error. Most of the missing values occurred because the player did not perform well in the mentioned area, such as batting, bowling, or T20 performance.

For instance, there were many bowlers whose batting performance was missing, whereas there were many batsmen whose bowling performance was missing. Also, there were many players whose T20 performance was missing, as they did not participate in the T20 matches.

Therefore, the missing values in the data set were mostly due to the lack of performance by the player, not the lack of information.

Handling Missing Values:

This is where imputation comes in, replacing the blank cells with the required values. Deletion is not an option since this would mean losing some information about the players involved.

Since the blank cells represent a lack of performance, which is usually zero, they were replaced with 0.

For the IPL batting columns such as Inns, Runs, BF, NO, 4s, 6s, 0s, 50s, 100s, the blank cells were replaced with 0 since they did not perform in any of these categories. The same applies to the blank cells in the columns for the average and strike rate since they did not get a chance to bat in any match.

For the T20 batting columns such as TMat, TInns, TRuns, TBF, TNO, T4s, T6s, T0s, T50s, T100s, the blank cells were replaced with 0 since they did not perform in any T20 match.

For the bowling columns such as B_Inns, B_Balls, B_Runs, B_Maidens, B_Wkts, B_4w, B_5w, the blank cells were replaced with 0 since they did not perform in any of these categories. The blank cells in the columns for B_Avg, B_Econ, B_SR were replaced with 0 since they did not get a chance to perform in any match.

For the T20 bowling columns such as B_TInns, B_TBalls, B_TRuns, B_TMaidens, B_TWkts, B_TAvg, B_TEcon, B_TSR, B_T4w, B_T5w, the blank cells were replaced with 0 since they did not get a chance to perform in any T20 match.

Data Cleaning:

The data set had cleaning processes carried out on it. This involved checking all the columns in the data set and replacing blank values with a zero where blank meant no performance. This ensured a complete data set for all the players.

Auditing of the numeric columns was done to ensure that all the values were in the correct format for computation.

For the SOLD_PRICE column, despite the absence of any missing values, the values were in text representation for crores and lakhs. These were later changed to a numeric representation in lakhs for ease of computation.

After the cleaning process, the data set had no missing values.

Conversion of String Values to Numeric Values:

Some of the values were originally stored as string values that needed to be converted to numeric values to allow the data to be used for computational purposes.

The values stored in the column named SOLD_PRICE were originally stored as string values where the prices were given in crores and lakhs, like 6CR, 2.60CR, and 50L. They were converted to numeric values and expressed in lakhs, considering that 1 crore is equivalent to 100 lakhs.

A new column named Team Code was added to the dataset to represent the teams using numeric labels from 0 to 9. Each of the teams was given a unique number using nominal encoding, considering that the teams were originally given names that were nominal and had no ordinal relationship with each other.

The original column named Team was left unchanged, while the column named Team Code was used for the purpose of analysis.

Q2. Player Distribution Analysis

Team Encoding:

In order to analyze the distribution of the total number of players among the teams, a new column was added, which was named Team Code. In this column, a numerical value was assigned to each team, ranging from 0 to 9.

The new column was used solely as a label to identify the different teams, without any other form of ranking.

A new sheet was added, which was named Legend, to show the correlation of the team names with their codes.

The new column, Team Code, was used to create the Pivot Tables, whereas the original column, Team was left unchanged.

Number of Players per Team:

The number of players per team was determined using a Pivot Table and then represented in a column chart. From the chart, it is clear that the number of players in each team is comparable.

Most teams have a fair number of players ranging from 17 to 21 players. None of the teams has a significantly high or low number of players compared to the others

Average Age by Team:

A Pivot Table was used to calculate the average age of the players in each team, using the Team Code and Age fields. The result was then used to create a column chart.

The chart shows that the average age of the players in the teams varies little. Most teams have an average age of 27-30 years, which means that there is a fair balance of young and experienced players.

No team has a notably higher or lower average age than the others.

Distribution of Playing Roles within Teams:

A pivot table has been created to analyze the distribution of playing roles within teams, using the columns ‘Team Code’ and ‘Playing Role.’ This information has been further represented in the form of a stacked column chart.

The chart illustrates the number of players in the Batting, Bowling, and All-rounder roles across teams.

From the chart, it is evident that teams have a mix of batting, bowling, and all-rounder players. The distribution is somewhat uniform across teams, although some teams have slightly more players in specific roles than others.

Overall Observation:

This analysis has shown that the distribution of the number of players in each team, the average age of the players, as well as the position of the player, have been fairly distributed among the teams.

Q3. Age Demographics analysis

Basic Age Statistics:

In order to determine the age distribution of the players, the Age variable was used to extract the following basic age statistics.

The mean of the ages of the players is 28, which shows that the majority of the players are located around the late twenties of their lives. The median of the ages of the players is 29, which shows that the distribution of the ages of the players is divided into two halves, with half of the players being younger than 29, and the other half of the players being older than 29. The mode of the ages of the players is also 29, which shows that the age of 29 is the most common among the ages of the players.

The standard deviation of the ages of the players is around 4.54, which shows that the majority of the ages of the players are located around the mean, with very little variation.

The minimum age of the players is 19, whereas the maximum age of the players is 41, showing a variation of 22 years among the ages of the players.

Age Group Distribution:
The participants were divided into five age groups: 20-25 years, 26-30 years, 31-35 years, 36-40 years, and 40+ years. The number of participants in each group was determined using a Pivot Table.

According to the above chart and table, the 26-30 years age group has the largest number of participants, totaling 61. The next age group is the 31-35 years group, which has 57 participants. The 20-25 years age group has 53 participants, which is also significant.

For the age groups above 35 years, the number of participants decreases significantly. The 36-40 years age group has 10 participants, and the 40+ years age group has 1 participant.

Most Represented Age Group:
The age group of 26-30 is the most represented age group, consisting of a total of 61 players.

This suggests that the majority of the players fall in the late twenties.

Q.4 Top Batsmen Identification Identify the top 10 batsmen based on: Total runs scored in IPL, Batting average (Avg) and Strike rate (SR) Create a comparative visualization and discuss which metric is most important for identifying the best batsmen.

- Insight: The analysis reveals a clear distinction between "Accumulators" and "Finishers." Players like Virat Kohli and Shikhar Dhawan lead in total volume, providing a reliable foundation for their teams. However, when filtered for impact, players like Tim David and Andre Russell emerge as leaders in Strike Rate (SR > 175).
- Metric Importance: While Total Runs indicate longevity, Strike Rate is the most critical metric for identifying the "best" T20 batsmen. In a limited-overs format, the ability to score rapidly (SR) often has a higher correlation with winning than simply maintaining a high average.

Q5. Consistency vs. Aggression (Quadrant Analysis)

I developed a scatter plot mapping Batting Average (X-axis) against Strike Rate (Y-axis) to categorize player archetypes based on the dataset's median values.

- Premium Players (High Avg, High SR): These are the elite "Match Winners" (e.g., K.L. Rahul, David Warner) who consistently score big at a high pace.
- Anchors (High Avg, Low SR): These players provide stability, often batting through the innings to prevent collapses, though they may face criticism for "slow" scoring in high-target chases.
- Aggressive/Inconsistent (Low Avg, High SR): These are "High-Risk" players who provide quick momentum but are prone to early dismissals.
- Discussion: The trade-off is central to team strategy; a balanced lineup requires "Anchors" to mitigate risk and "Aggressive" players to exploit powerplays and death overs.

Q6. Boundary Analysis

I analyzed the scoring patterns by calculating the ratio of 4s to 6s and the correlation between boundaries and total runs.

- Key Finding: There is a near-perfect correlation (approx. 0.99) between Total Boundaries (4s + 6s) and Total Runs. This confirms that the ability to find the rope is the primary driver of high scores in the IPL.
- Player Styles: * 4s Reliant: Players with a high 4:6 ratio (technical batsmen) rely on gap-finding and placement.
    - 6s Reliant: Players with a low ratio (power hitters) rely on raw strength and clearing the boundary.
- Conclusion: Teams with a higher "Boundary Contribution Percentage" (runs from boundaries vs. total runs) are statistically more likely to win matches.

Q7. IPL vs. T20 Performance Comparison

This analysis compared individual performance in the IPL against the players' overall T20 career statistics to identify "League Specialists."

- IPL Over-performers: I identified players whose IPL Strike Rate is significantly higher (Diff > 15) than their career T20 average. This suggests they are better suited to Indian conditions or peak during the IPL window.
- Under-performers: Conversely, some international stars show a dip in IPL performance, likely due to the higher quality of bowling or the pressure of the price tag.
- Correlation: The correlation between IPL runs and Career T20 runs (approx. 0.73) suggests that while general T20 form is a good predictor, the IPL remains a unique environment where specific "sub-continent specialists" thrive more than they do globally.

Q8. Elite Bowlers Identification

The performance of bowlers was analyzed using four key metrics: Total Wickets (B_Wkts), Bowling Average (B_Avg), Economy Rate (B_Econ), and Bowling Strike Rate (B_SR). Each bowler was ranked separately based on these metrics to evaluate different aspects of bowling performance such as wicket-taking ability, efficiency, and run control.

A composite ranking score was calculated by taking the average of the four individual rankings. This helped combine all the performance indicators into a single value to identify the most effective bowlers overall.

The analysis revealed that bowlers who consistently performed well across all four metrics ranked higher in the composite ranking. These bowlers can be considered the elite bowlers in the dataset as they demonstrate a balanced combination of wicket-taking ability and effective run control.

Q9. Economy vs Wicket-Taking Analysis

To analyze the relationship between economy rate and wicket-taking ability, a scatter plot was created using Economy Rate (B_Econ) on the X-axis and Total Wickets (B_Wkts) on the Y-axis. Each point in the chart represents a bowler.

Based on their position in the scatter plot, bowlers were categorized into three groups. Death bowlers have high wickets but relatively higher economy rates, indicating they often bowl in high-pressure situations such as the final overs. Economical bowlers have lower economy rates but fewer wickets, showing strong run control. Strike bowlers maintain a balance between wicket-taking ability and controlled economy rates.

The visualization helps identify different bowling styles and highlights bowlers who are effective both in taking wickets and controlling runs.

Q10. Bowling Impact Analysis

The Bowling Impact Score was calculated using wickets, economy rate, and wicket haul performances. Bowlers were ranked based on this score and compared with their auction prices. The analysis shows that some bowlers with very high impact scores, such as Tim Southee and Mustafizur Rahman, were sold at relatively lower prices, indicating potential undervaluation. Conversely, some bowlers such as Lockie Ferguson and Avesh Khan were sold at higher prices despite having lower impact scores compared to other bowlers. This suggests that auction prices are influenced not only by statistical performance but also by factors such as player reputation, experience, and team demand.

Q11 All-Rounder Performance Analysis

To identify the most balanced all-rounders, T20 performance statistics were analyzed. The evaluation considered both batting and bowling contributions using the following metrics:

- Batting: T20 Runs (TRuns), Batting Average (TAvg), Strike Rate (TSR)

- Bowling: T20 Wickets (B_TWkts), Bowling Average (B_TAvg), Economy Rate (B_TEcon)

A Balanced Score was created to combine both batting and bowling impact into a single performance indicator. This score weighted batting consistency, strike efficiency, wicket-taking ability, and economy control.

After sorting players based on the Balanced Score, Rashid Khan emerged as the most balanced all-rounder. His strong wicket-taking record, excellent economy rate, and effective lower-order batting contribution make him a high-impact dual performer.

Other players such as Chris Jordan and Mitchell Santner also ranked highly, showing consistent performance in both departments. Players who had strong batting but weak bowling (or vice versa) ranked lower, as true balance requires contribution in both areas.

Overall, the analysis demonstrates that bowling effectiveness (wickets and economy) plays a significant role in determining an all-rounder's overall balance in T20 cricket.

Q12 Value All-Rounder Analysis

To evaluate value for money, an All-Rounder Score (AR Score) was calculated using the formula:

AR Score=(TRuns/10)+(TSR/2)+(BTWkts×20)−(BTEcon×10)\text{AR Score} = (TRuns/10) + (TSR/2) + (B_TWkts × 20) - (B_TEcon × 10)

Score=(TRuns/10)+(TSR/2)+(BTWkts×20)−(BTEcon×10)

This formula combines batting output, scoring efficiency, wicket-taking ability, and bowling economy into a single performance metric.

The players were then ranked based on AR Score to identify the top 10 performing all-rounders.

To assess cost efficiency, a Value Index was calculated:

Value Index= AR Score ÷ Sold Price

Players with a high AR Score but relatively moderate or lower price were identified as value picks, as they deliver strong performance per unit cost.

The analysis revealed that while premium players such as Rashid Khan justify their higher price through consistent high performance, some mid-priced players offer excellent returns relative to their cost. These players represent smart investments for franchises seeking balanced performance within budget constraints.

This evaluation highlights that performance alone does not determine value, price efficiency plays an equally important role in strategic team building.

Based on the Value Index analysis, Dwaine Pretorius emerges as the most cost efficient all-rounder, delivering strong performance relative to his low price. While players like Mitchell Santner and Chris Jordan also provide high impact, their higher prices reduce overall value efficiency. This analysis highlights the importance of balancing performance output with financial investment when selecting all-rounders.

Q13 – Price Distribution Analysis

The distribution of player auction prices (converted into crores) was analyzed using a histogram and percentile measures. The histogram indicates a positively skewed distribution, where the majority of players fall within the lower price brackets. Most players are concentrated in the 0.2 to 3 crore range, while only a small proportion of players command premium prices above 10 crores.

The percentile analysis further supports this observation. The 25th percentile value is approximately 0.66 crores, indicating that 25% of players are priced below this amount. The median (50th percentile) price is around 2.7 crores, meaning half of the players are sold below this value. The 75th percentile is 7.75 crores, and the 90th percentile is 10.75 crores, showing that only a limited number of players belong to the high-price segment.

Additionally, the comparison of average prices across playing roles reveals that all-rounders have the highest average price (approximately 5.18 crores), followed closely by batsmen (5.05 crores), while bowlers have a comparatively lower average price (3.09 crores). This suggests that franchises place greater financial value on versatile players who contribute in multiple departments.

Overall, the analysis shows that auction investments are largely concentrated in low-to-mid price ranges, with only a few elite players commanding exceptionally high bids.

Q14 – Performance vs Price Correlation Analysis

To evaluate the relationship between player performance and auction price, correlation analysis was conducted separately for batsmen, bowlers, and all-rounders.

For batsmen, the correlation between price and batting metrics such as runs, batting average, strike rate, and milestone performances (50s and 100s) ranged between 0.29 and 0.48. These values indicate a moderate positive relationship, suggesting that stronger batting performance is associated with higher auction prices.

For bowlers, the correlation between price and bowling metrics (wickets, economy rate, and bowling average) was relatively weak, ranging from 0.06 to 0.21. This implies that bowling performance alone does not strongly determine player valuation in the auction.

For all-rounders, the correlation between price and the combined All-Rounder Score was approximately 0.16, indicating a weak positive relationship. This suggests that while performance contributes to pricing decisions, auction prices are influenced by additional factors such as player reputation, experience, brand value, and recent form.

Overall, the results indicate that batting performance has a stronger influence on auction price compared to bowling performance, though statistical performance alone does not fully explain player valuation.

Q15 – Undervalued and Overvalued Players (Regression Analysis)

A simple linear regression model was developed to predict player price using the All-Rounder (AR) Score. The regression equation obtained was:

Price = 5.17771 + 0.0000658 * AR Score

The positive slope indicates that higher AR Scores are associated with higher predicted prices, although the magnitude of the coefficient suggests that the relationship is relatively weak.

Predicted prices were calculated for all all-rounders and compared with their actual auction prices. The difference between predicted and actual prices was used to classify players as undervalued or overvalued.

Players whose predicted price exceeded their actual auction price were identified as undervalued, indicating that their statistical performance suggests they should have been sold at a higher price. Conversely, players whose actual price exceeded the predicted value were classified as overvalued, suggesting that non-performance factors may have influenced their higher auction valuation.

The regression analysis demonstrates that while performance metrics contribute to price determination, auction dynamics are also significantly influenced by market perception, brand image, international exposure, and team strategy considerations.

Q20. Predictive Model for Player Success

To analyze player success and predict auction price, both classification and regression approaches were used based on player performance statistics.

Classification Model

Players were first classified as high-value or low-value based on the median auction price. The median price of all players in the dataset was calculated as 2.7 crores. Players whose auction price was greater than the median were classified as High Value, while those with prices less than or equal to the median were classified as Low Value. This classification helps identify players who command higher market value relative to the majority of players in the auction.

Regression Model

A regression model was developed to predict player auction price using performance metrics. Runs scored were initially used as the primary predictor variable. The regression equation obtained was:

Price = 3.409411 + 0.003044 * (Runs)

The positive slope indicates that higher run totals are associated with higher auction prices. Specifically, for every additional run scored, the predicted auction price increases by approximately 0.003 crores, suggesting a positive relationship between batting performance and player valuation.

Model Evaluation

The regression model was evaluated using RMSE (Root Mean Squared Error) and R^2 values. The calculated RMSE was approximately 3.85 crores, indicating the average difference between the predicted price and the actual auction price. This suggests that while the model captures some relationship between performance and price, there are still other factors influencing player valuation.

To determine which player statistics best predict auction price, the coefficient of determination (R^2) was calculated for multiple performance variables.

- Runs: RSQ = 0.2318
- Batting Average: RSQ = 0.2200
- Strike Rate: RSQ = 0.2184
- Number of 100s: RSQ = 0.0862
- Matches Played: RSQ = 0.2755

Among these variables, matches played shows the highest explanatory power with an R^2 value of approximately 0.276, indicating that player experience has the strongest relationship with auction price in this dataset. Runs, batting average, and strike rate also show moderate relationships, while the number of centuries appears to have a weaker influence.

Hypothesis Testing

Several hypotheses were formulated to test the relationship between player performance and auction price.

General Hypothesis 1
Null Hypothesis (H0): Players age, runs, and number of centuries do not significantly influence player price.
Alternative Hypothesis (H1): Players age, runs, and number of centuries significantly influence player price.

General Hypothesis 2
Null Hypothesis (H0): Cap experience, playing role, and number of matches played do not significantly influence player price.

Alternative Hypothesis (H1): Cap experience, playing role, and number of matches played significantly influence player price.

Additional Hypotheses

H3
H0: Strike rate has no impact on player auction price.
H1: Strike rate significantly influences player auction price.

H4
H0: The number of wickets taken does not affect player price.
H1: Players with more wickets tend to receive higher auction prices.

H5
H0: Playing role has no impact on auction price.
H1: All-rounders tend to receive higher auction prices compared to specialist players.

H6
H0: Batting average does not affect player auction price.
H1: Higher batting averages lead to higher auction prices.

H7
H0: The number of matches played does not influence player price.
H1: Players with more match experience receive higher auction prices.

Conclusion

The predictive analysis indicates that player performance metrics such as runs, batting average, and strike rate have a moderate relationship with auction price. However, the number of matches played appears to have the strongest influence among the variables analyzed, suggesting that player experience plays a significant role in determining auction value. Despite these relationships, the moderate R^2 values indicate that additional factors such as player reputation, team strategy, and recent performance also contribute to the final auction price.

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.534445783
R Square	0.285632295
Adjusted R Square	0.263852792
Standard Error	3.875530656
Observations	170

ANOVA

	df	*SS*	*MS*	*F*	*Significance F*
Regression	5	984.8990045	196.9798009	13.11472961	9.58759E-11
Residual	164	2463.23701	15.01973787		
Total	169	3448.136015			

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*Lower 95.0%*	*Upper 95.0%*
Intercept	2.752750961	0.389746549	7.062925809	4.40724E-11	1.983182924	3.522318998	1.983182924	3.522318998
42	0.002569198	0.00152679	1.68274435	0.094327425	-0.000445503	0.005583899	-0.000445503	0.005583899
10.5	-0.021021276	0.050063019	-0.419896295	0.675110793	-0.119872437	0.077829885	-0.119872437	0.077829885
84	0.019503958	0.008312038	2.346471369	0.020147355	0.003091553	0.035916364	0.003091553	0.035916364
0	-0.055528523	0.176619689	-0.314395999	0.753619895	-0.404270206	0.293213159	-0.404270206	0.293213159
0	0.000913926	0.719342337	0.001270502	0.99898783	-1.419452389	1.421280241	-1.419452389	1.421280241

Regression for Batting

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.29491447
R Square	0.08697454
Adjusted R Squar	0.05896763
Standard Error	4.25365776
Observations	169

ANOVA

	df	*SS*	*MS*	*F*	*Significance F*
Regression	5	280.9454215	56.1890843	3.105466619	0.010506071
Residual	163	2949.257508	18.09360434		
Total	168	3230.202929			

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*Lower 95.0%*	*Upper 95.0%*
Intercept	3.63899679	0.412102215	8.830325705	1.5757E-15	2.825249605	4.452743967	2.825249605	4.452743967
32	0.04194541	0.023996907	1.747950902	0.082355457	-0.005439468	0.089330298	-0.005439468	0.089330298
28.5	0.0144679	0.024710203	0.585502947	0.559019327	-0.034325479	0.063261273	-0.034325479	0.063261273
6.8	0.00048859	0.00053294	0.916788139	0.360608105	-0.000563763	0.00154095	-0.000563763	0.00154095
0	0.6541775	0.742242961	0.881352245	0.37942495	-0.811473725	2.119828725	-0.811473725	2.119828725
0	-1.9774407	1.551893897	-1.274211291	0.204402228	-5.041848625	1.086967173	-5.041848625	1.086967173

Regression for Bowling

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.62002073
R Square	0.384425705
Adjusted R Square	0.369411698
Standard Error	3.482035683
Observations	169

ANOVA

	df	*SS*	*MS*	*F*	*Significance F*
Regression	4	1241.77304	310.4432599	25.60447059	1.71076E-16
Residual	164	1988.429889	12.1245725		
Total	168	3230.202929			

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*Lower 95.0%*	*Upper 95.0%*
Intercept	2.106674172	0.366811138	5.743212115	4.40072E-08	1.382392897	2.830955447	1.382392897	2.830955447
0	0.002255112	0.000483328	4.665802633	6.35029E-06	0.001300765	0.00320946	0.001300765	0.00320946
0	0.016954292	0.00505615	3.353202164	0.000992045	0.00697075	0.026937834	0.00697075	0.026937834
32	0.04919308	0.010884152	4.519698142	1.18079E-05	0.027701946	0.070684214	0.027701946	0.070684214
6.8	0.00032488	0.000442931	0.733476962	0.464314778	-0.000549703	0.001199464	-0.000549703	0.001199464

Regression for All rounders

To understand how player performance influences auction prices, regression models were developed separately for batsmen, bowlers, and all-rounders using relevant performance metrics.

Regression Analysis for Batsmen

A regression model was developed using batting statistics such as total runs scored, batting average, strike rate, number of half-centuries, and centuries to predict player auction price. The results show that batting performance has a noticeable influence on player valuation. Players who score higher runs and maintain strong batting averages tend to attract higher auction prices. Strike rate and milestone performances such as 50s and 100s also contribute to price determination, as they indicate consistency and match-winning ability. Overall, the regression analysis suggests that batting performance is one of the key factors influencing the auction value of specialist batsmen.

Regression Analysis for Bowlers

For bowlers, the regression model included bowling statistics such as wickets taken, economy rate, bowling average, and wicket haul performances (four-wicket and five-wicket hauls). The results indicate that wickets taken and economy rate are important indicators of bowler performance and contribute to price prediction. Bowlers who consistently take wickets and maintain a low economy rate are generally valued more highly in auctions. However, the relationship between bowling metrics and price appears weaker compared to batting metrics, suggesting that other factors such as experience, reputation, and team strategy also influence the valuation of bowlers.

Regression Analysis for All-Rounders

All-rounders contribute in both batting and bowling departments, making them highly valuable assets for teams. The regression model for all-rounders included both batting metrics (runs and strike rate) and bowling metrics (wickets and economy rate). The results indicate that players who perform well in both departments tend to command higher auction prices. Teams value all-rounders because they provide flexibility in team composition and contribute to multiple aspects of the game. Consequently, strong overall performance in both batting and bowling significantly increases the likelihood of higher auction prices.

Overall Insight

The regression analysis demonstrates that player performance metrics do influence auction prices, with batting performance showing a relatively stronger relationship compared to bowling metrics. All-rounders tend to have higher valuation potential due to their dual contribution. However, the moderate explanatory power of the regression models suggests that auction prices are also affected by additional factors such as player reputation, international experience, recent form, and team requirements.